

Recall from previously:

- A set of vectors $\{u_1, \dots, u_p\}$ in $\mathbb{R}^n$ is **orthogonal** if $u_k \cdot u_j = 0$ for all $j \neq k$.
- **Theorem 6.3** Let A be an $m \times n$ matrix, then

$$(\text{Row}(A))^\perp = \text{Null}(A) \text{ and } (\text{Col}(A))^\perp = \text{Null}(A^T).$$

- **Theorem 6.4:** If $S = \{u_1, \dots, u_p\}$ is an orthogonal set of non-zero vectors in $\mathbb{R}^n$, then S is linearly independent and hence a basis for $\text{Span } S$.
- An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W that is also an orthogonal set.
- **Theorem 6.5:** Let $\{u_1, \dots, u_p\}$ be an orthogonal basis for a subspace W of $\mathbb{R}^n$. For each $y \in W$, then

$$y = c_1 u_1 + \dots + c_p u_p$$

where

$$c_j = \frac{y \cdot u_j}{u_j \cdot u_j}.$$

- Let u be a non-zero vector in $\mathbb{R}^n$ and let v be a vector in $\mathbb{R}^n$, then the **orthogonal projection** of v onto u , denoted by $\text{proj}_u v$, is given by

$$\text{proj}_u v = \frac{v \cdot u}{u \cdot u} u$$

Let $L = \text{Span}\{u\}$, then we also write $\text{proj}_L v$ for the orthogonal projection of v onto the line defined by the span of u ; namely

$$\text{proj}_L v = \text{proj}_u v.$$

- **Note:** Theorem 6.5 says that

$$y = \text{proj}_{u_1} y + \text{proj}_{u_2} y + \dots + \text{proj}_{u_p} y.$$

- A set $\{u_1, \dots, u_p\}$ is **orthonormal** if it is orthogonal and $\|u_k\| = 1$ for all k .
- **Theorem 6.6:** An $m \times n$ matrix U has orthonormal columns if and only if $U^T U = I_n$.
- **Theorem 6.7:** Let U be an $m \times n$ matrix with orthonormal columns. Let $x, y \in \mathbb{R}^n$. Then
 1. $\|Ux\| = \|x\|$.
 2. $(Ux) \cdot (Uy) = x \cdot y$.
 3. $(Ux) \cdot (Uy) = 0$ if and only if $x \cdot y = 0$.
- An **orthogonal matrix** U is a square matrix where $U^{-1} = U^T$.

Orthogonal Decomposition Theorem: Let W be a subspace of $\mathbb{R}^n$. Then each $y \in \mathbb{R}^n$ can be uniquely written as

$$y = \hat{y} + z$$

where $\hat{y} \in W$ and $z \in W^\perp$.

Moreover, if $\{u_1, \dots, u_p\}$ is any orthogonal basis for W , then

$$\hat{y} = \left(\frac{y \cdot u_1}{u_1 \cdot u_1} \right) u_1 + \dots + \left(\frac{y \cdot u_p}{u_p \cdot u_p} \right) u_p,$$

and $z = y - \hat{y}$.

Definition 1. The vector $\hat{y}$ is the **orthogonal projection** of y onto the subspace W . Written as

$$\hat{y} = \text{proj}_W y.$$

1. Prove the Theorem.

Best Approximation Theorem: Let W be a subspace of $\mathbb{R}^n$. Let $\hat{y}$ be the orthogonal projection of y onto W , then

$$\|y - \hat{y}\| < \|y - v\|$$

for all $v \in W \setminus \{\hat{y}\}$.

2. Prove the Theorem.

Theorem 6.10: If $\{u_1, \dots, u_p\}$ is an orthonormal basis for W , then

$$\text{proj}_W y = (y \cdot u_1)u_1 + \cdots + (y \cdot u_p)u_p.$$

If $U = [u_1 \ \dots \ u_p]$, then

$$\text{proj}_W y = UU^T y.$$

3. Prove the theorem.

Question: How can we find an orthogonal basis for a given subspace?

4. Let

$$W = \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right\}.$$

Find an orthogonal basis for W .

Gram-Schmidt Orthogonalization. Given a basis $\{x_1, \dots, x_p\}$ for a non-zero subspace W of $\mathbb{R}^n$, define

$$\begin{aligned} v_1 &= x_1 \\ v_2 &= x_2 - \left(\frac{x_2 \cdot v_1}{v_1 \cdot v_1} \right) v_1 \\ v_3 &= x_3 - \left(\frac{x_3 \cdot v_1}{v_1 \cdot v_1} \right) v_1 - \left(\frac{x_3 \cdot v_2}{v_2 \cdot v_2} \right) v_2 \\ &\vdots \\ v_p &= x_p - \left(\frac{x_p \cdot v_1}{v_1 \cdot v_1} \right) v_1 - \dots - \left(\frac{x_p \cdot v_{p-1}}{v_{p-1} \cdot v_{p-1}} \right) v_{p-1}. \end{aligned}$$

Then $\{v_1, \dots, v_p\}$ is an orthogonal basis for W . In addition $\text{Span}\{v_1, \dots, v_k\} = \text{Span}\{x_1, \dots, x_k\}$ for all $1 \leq k \leq p$.

5. Prove the Theorem.

QR Factorization. If A is an $m \times n$ matrix with linearly independent columns, then $A = QR$ where Q is an $m \times n$ matrix whose columns form an orthonormal basis for the column space of A and R is an upper triangular $n \times n$ matrix with positive entries along the diagonal.

6. Prove the Theorem.

7. Let $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$. Find QR factorization of A .